Day 12 Notes

NLP and Text Processing

What is NLP?

NLP (Natural Language Processing) is a branch of AI that enables computers to understand, interpret, and generate human language.

It combines:

- Linguistics
- Machine Learning
- Artificial Intelligence

Applications of NLP:

- Chatbots
- Sentiment Analysis (Positive/Negative detection)
- Translation
- Voice assistants
- Text summarization
- Spam detection

Positive & Negative Sentences (Sentiment Analysis)

Sentiment Analysis is used to determine whether a sentence is:

Positive, Negative or Neutral.

Examples:

Sentence	Sentiment
"I love this product."	Positive
"This service is terrible."	Negative
"The movie was okay."	Neutral

Used in:

- Product reviews
- Social media analysis
- Customer feedback systems

NLTK – Natural Language Toolkit

1. What is NLTK?

NLTK (Natural Language Toolkit) is a Python library used for:

- Text processing
- Tokenization
- Stop word removal
- Stemming
- Lemmatization
- POS tagging
- Named Entity Recognition (NER)

It helps interpret and analyze human language data.

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2. Installation & Setup

```
pip install nltk          # Install NLTK
import nltk               # Import library
nltk.download('all')      # Download all NLTK packages
```

---

3. Main Features of NLTK (7 Core Features)

1. Tokenization

Breaking text into smaller parts (tokens).

Example:

Sentence:

NLP is interesting

Output:

```
["NLP", "is", "interesting"]
```

Code:

```
from nltk.tokenize import word_tokenize

text = "NLP is interesting"
tokens = word_tokenize(text)
print(tokens)
```

---

2. Stop Word Removal

Stop words are common words that do not add much meaning.

Examples:

- is
- the
- and
- in
- of

Code:

```
from nltk.corpus import stopwords
```

#Removes unnecessary words to improve analysis.

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3. Stemming

Reduces a word to its root form (not always meaningful).

Example:

- Steadily → Steadi
- Playing → Play
- Running → Run

Stemming may not produce proper English words.

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4. Lemmatization

Converts words to meaningful base form.

Example:

- Running → Run
- Better → Good
- Steadily → Steady

Code:

```
from nltk.stem import WordNetLemmatizer
```

Lemmatization is more accurate than stemming.

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5. POS Tagging (Part of Speech Tagging)

Identifies the grammatical role of words.

POS = Part of Speech

Examples:

- Noun
- Verb
- Adjective
- Adverb

Code:

```
from nltk.tokenize import word_tokenize
import nltk

text = "NLP is very interesting"
tokens = word_tokenize(text)
pos_tags = nltk.pos_tag(tokens)
print(pos_tags)
```

Example Output:

```
[('NLP', 'NNP'), ('is', 'VBZ'), ('very', 'RB'), ('interesting',
'JJ')]
```

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6. NER – Named Entity Recognition

Identifies names of:

- People
- Organizations
- Locations
- Dates

Required Downloads:

```
nltk.download('words')
nltk.download('maxent_ne_chunker')
```

Used to extract:

- Company names
- Person names
- Places

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7. Text Processing & Text Preprocessing

Text Processing

Raw handling of text data.

Text Preprocessing

Cleaning and preparing text before training a model.

Steps:

1. Lowercasing
2. Removing punctuation
3. Removing stopwords
4. Tokenization
5. Stemming/Lemmatization

Preprocessing improves model accuracy.

- Invoice digitization
- Receipt scanning
- ID card reading
- Bank document extraction

Assignment- 4
Working with NLTK Codes

nltk_codes.py 5

assignment_4 > nltk_codes.py > ...

```python
# Assignment 4 – NLP using NLTK

import nltk
import string
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer
from nltk import pos_tag, ne_chunk

# Download required resources (run once)
nltk.download('punkt_tab')
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('maxent_ne_chunker')
nltk.download('words')

# Sample Text
text = """
Apple is a big technology company based in California.
Tim Cook is the CEO of Apple.
NLP is very interesting and useful in Artificial Intelligence.
"""

print("Original Text:\n", text)

# ----------------------------------------------
# 1. Lowercasing
text_lower = text.lower()
print("\nLowercased Text:\n", text_lower)

# ----------------------------------------------
# 2. Remove Punctuation
text_no_punct = text_lower.translate(str.maketrans('', '', string.punctuation))
print("\nText without Punctuation:\n", text_no_punct)

```

```python
# ----------------------------------------------
# 2. Remove Punctuation
text_no_punct = text_lower.translate(str.maketrans('', '', string.punctuation))
print("\nText without Punctuation:\n", text_no_punct)

# ----------------------------------------------
# 3. Tokenization
tokens = word_tokenize(text_no_punct)
print("\nTokens:\n", tokens)

# ----------------------------------------------
# 4. Stop Word Removal
stop_words = set[Any](stopwords.words('english'))
filtered_words = [word for word in tokens if word not in stop_words]
print("\nAfter Stopword Removal:\n", filtered_words)

# ----------------------------------------------
# 5. Stemming
stemmer = PorterStemmer()
stemmed_words = [stemmer.stem(word) for word in filtered_words]
print("\nStemmed Words:\n", stemmed_words)

# ----------------------------------------------
# 6. Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]
print("\nLemmatized Words:\n", lemmatized_words)

# ----------------------------------------------
# 7. POS Tagging
pos_tags = pos_tag(tokens)
print("\nPOS Tags:\n", pos_tags)
```

Output:

```
Problems 5   Output   Debug Console   Terminal   Ports

Original Text:

Apple is a big technology company based in California.
Tim Cook is the CEO of Apple.
NLP is very interesting and useful in Artificial Intelligence.


Lowercased Text:

apple is a big technology company based in california.
tim cook is the ceo of apple.
nlp is very interesting and useful in artificial intelligence.


Text without Punctuation:

apple is a big technology company based in california
tim cook is the ceo of apple
nlp is very interesting and useful in artificial intelligence


Tokens:
 ['apple', 'is', 'a', 'big', 'technology', 'company', 'based', 'in', 'california', 'tim', 'cook', 'is', 'the', 'ceo', 'of', 'apple', 'nlp', 'is',
'very', 'interesting', 'and', 'useful', 'in', 'artificial', 'intelligence']

After Stopword Removal:
 ['apple', 'big', 'technology', 'company', 'based', 'california', 'tim', 'cook', 'ceo', 'apple', 'nlp', 'interesting', 'useful', 'artificial', 'in
telligence']

Stemmed Words:
 ['appl', 'big', 'technolog', 'compani', 'base', 'california', 'tim', 'cook', 'ceo', 'appl', 'nlp', 'interest', 'use', 'artifici', 'intellig']

Lemmatized Words:
 ['apple', 'big', 'technology', 'company', 'based', 'california', 'tim', 'cook', 'ceo', 'apple', 'nlp', 'interesting', 'useful', 'artificial', 'in
telligence']

POS Tags:
 [('apple', 'NN'), ('is', 'VBZ'), ('a', 'DT'), ('big', 'JJ'), ('technology', 'NN'), ('company', 'NN'), ('based', 'VBN'), ('in', 'IN'), ('californi
a', 'NN'), ('tim', 'NN'), ('cook', 'NN'), ('is', 'VBZ'), ('the', 'DT'), ('ceo', 'NN'), ('of', 'IN'), ('apple', 'NN'), ('nlp', 'NN'), ('is', 'VBZ')
, ('very', 'RB'), ('interesting', 'JJ'), ('and', 'CC'), ('useful', 'JJ'), ('in', 'IN'), ('artificial', 'JJ'), ('intelligence', 'NN')]
sriyamk@Sriyas-MacBook-Pro assignments %
```